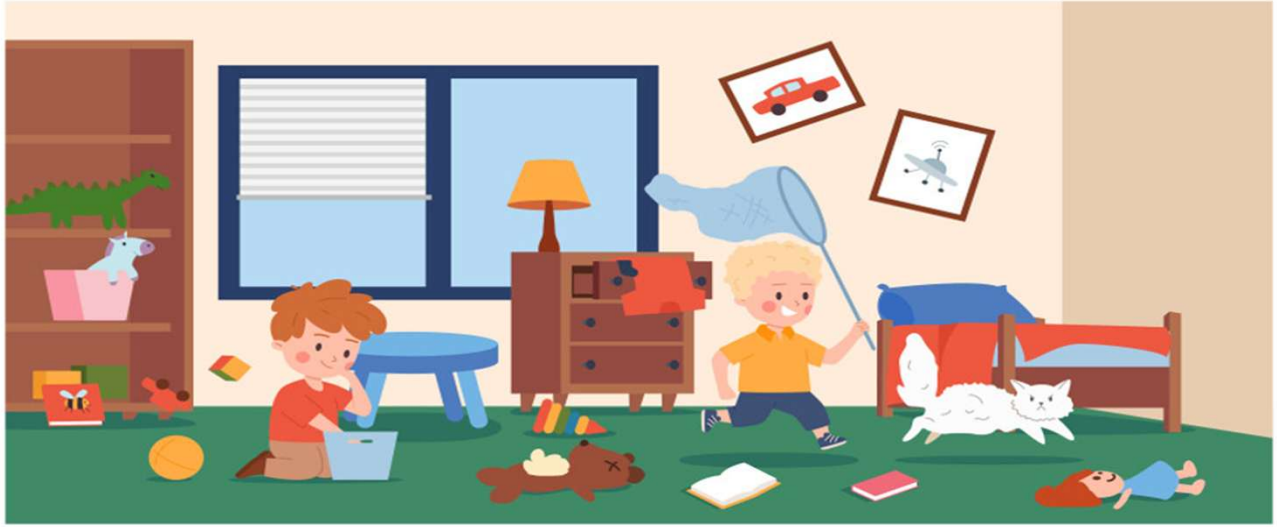
Classroom Activity 1

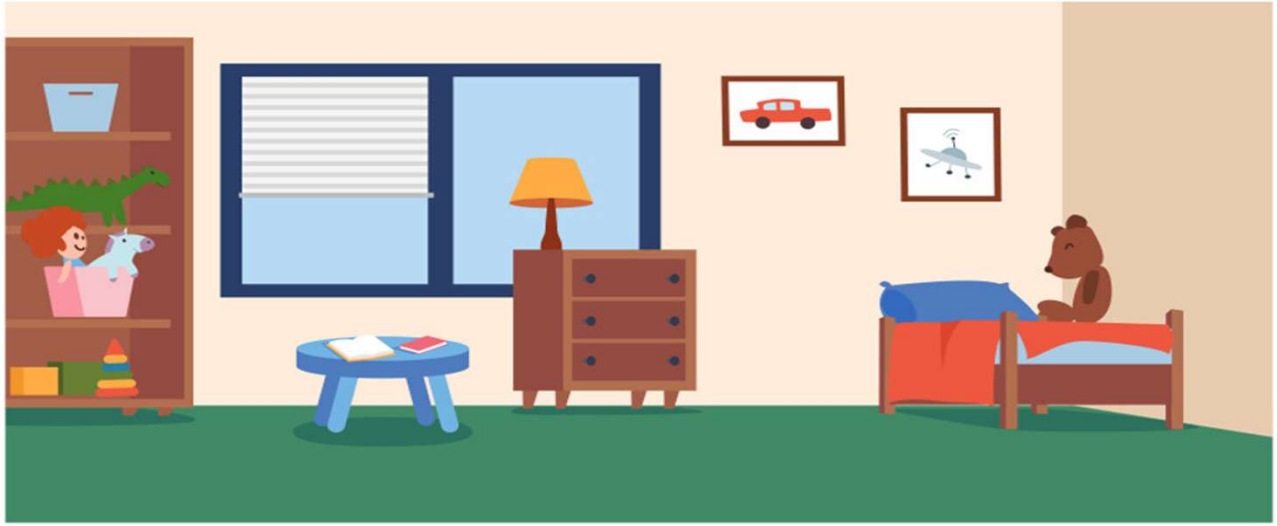
Topic: Entropy in Action

Date: 18.07.2025

Hours: 1 & 2

Entropy Example: A Child's Room





Entropy Example: A Messy Room

Hot Soup in a Cold Room – Entropy in Action (1/3)

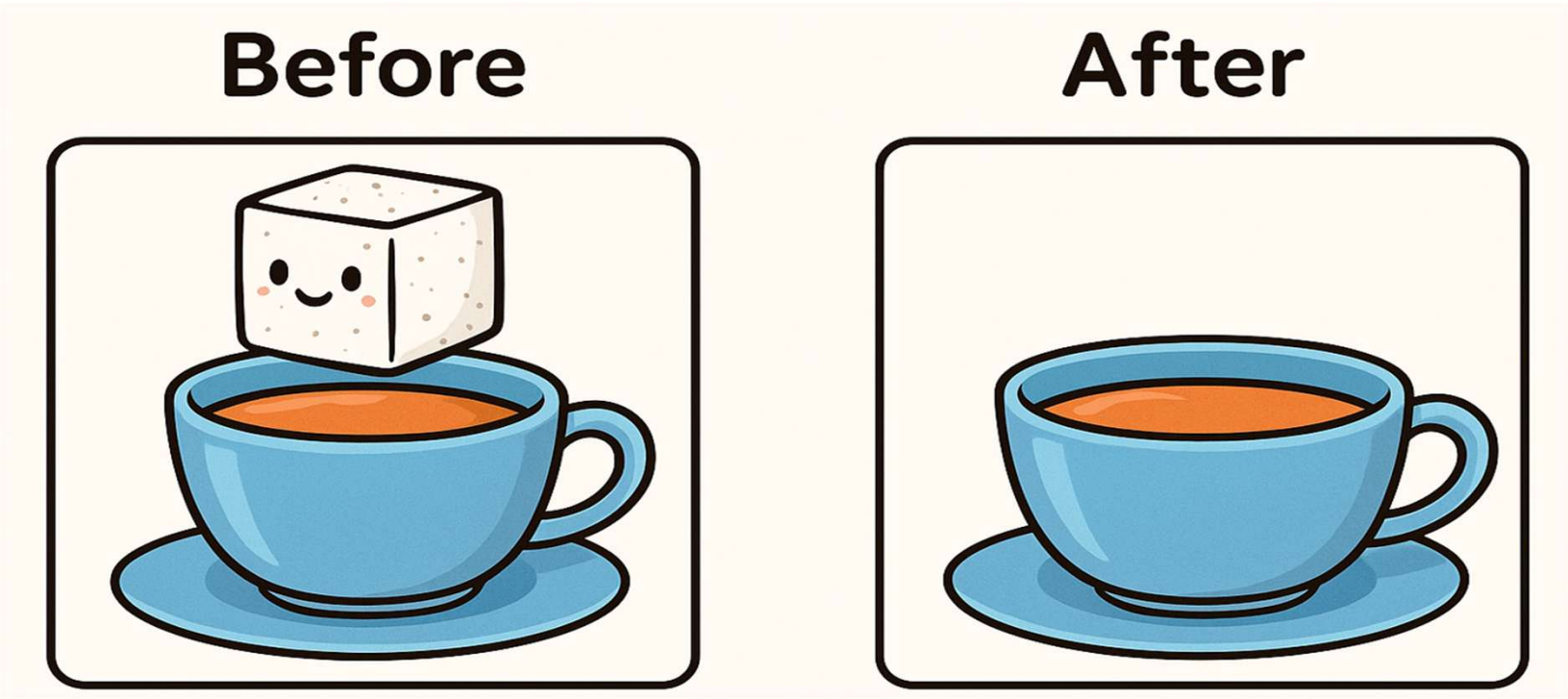
- Imagine you place a **steaming bowl of hot soup** on a table in a cold room.
  - Initially, the **soup is hot** (high temperature), and the **room is cold** (lower temperature).
  - Over time, the heat from the soup **spreads into the surrounding air**.
  - Eventually, the soup and the room air reach the **same temperature**—a state of **thermal equilibrium**.
- In this process:
  - The **heat energy spreads out**, becoming more disordered.
  - The **entropy of the system increases** because energy is no longer concentrated in the soup but is **dispersed more randomly** in the air molecules.

Hot Soup in a Cold Room – Entropy in Action (2/3)

Hot Soup in a Cold Room – Entropy in Action (3/3)

- According to thermodynamics:
  - **Entropy increases when energy becomes more evenly distributed across a system.**
  - If no work is done (like using a thermos), this process is irreversible—you can't get the soup hot again without adding energy from outside.

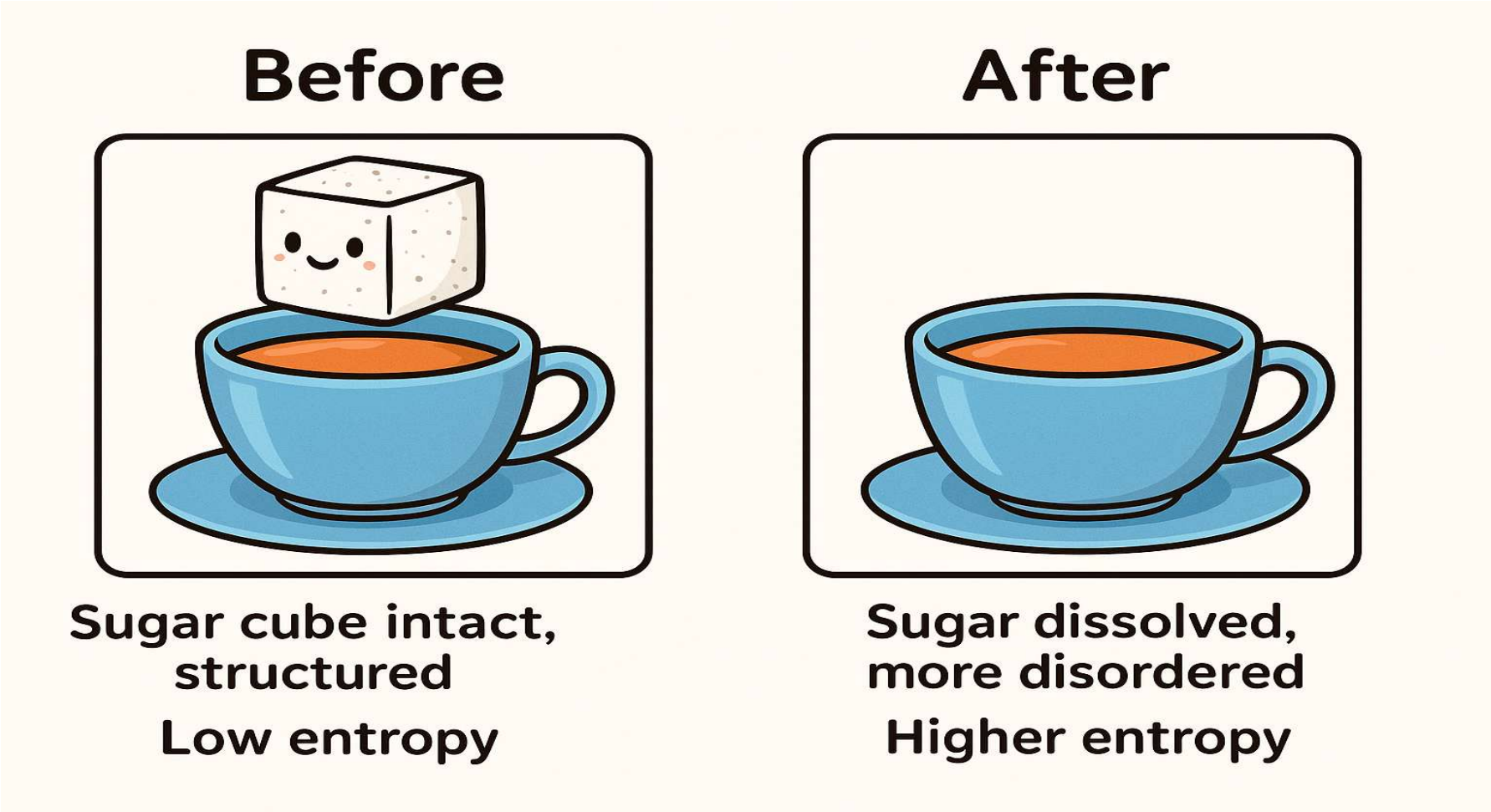
Sugar Cube and Entropy: A Simple Analogy (1/3)



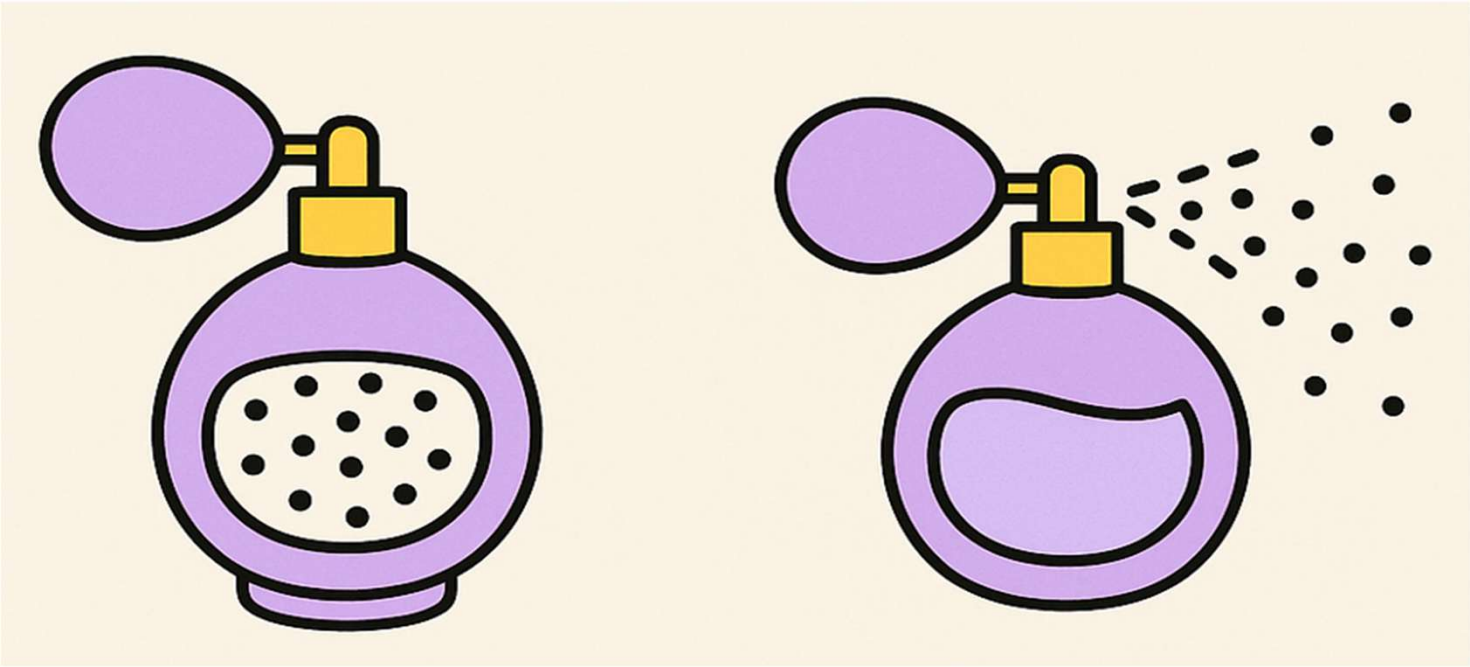
Sugar Cube and Entropy: A Simple Analogy (2/3)

- **Scenario 1: A Whole Sugar Cube**
  - Imagine you place a **sugar cube** gently at the bottom of a cup of tea, without stirring.
  - The cube is **compact and orderly**, with sugar molecules neatly stacked.
  - **Entropy is low** — the system (tea + sugar cube) is in an ordered state.
- **Scenario 2: Sugar Dissolving in Tea**
  - As time passes, the sugar cube **dissolves**, and its molecules **spread out** through the tea.
  - The molecules go from a **highly ordered state** (solid cube) to a **disordered state** (evenly mixed in the liquid).
  - Now, **entropy has increased** — the system is more **random** and **dispersed**.

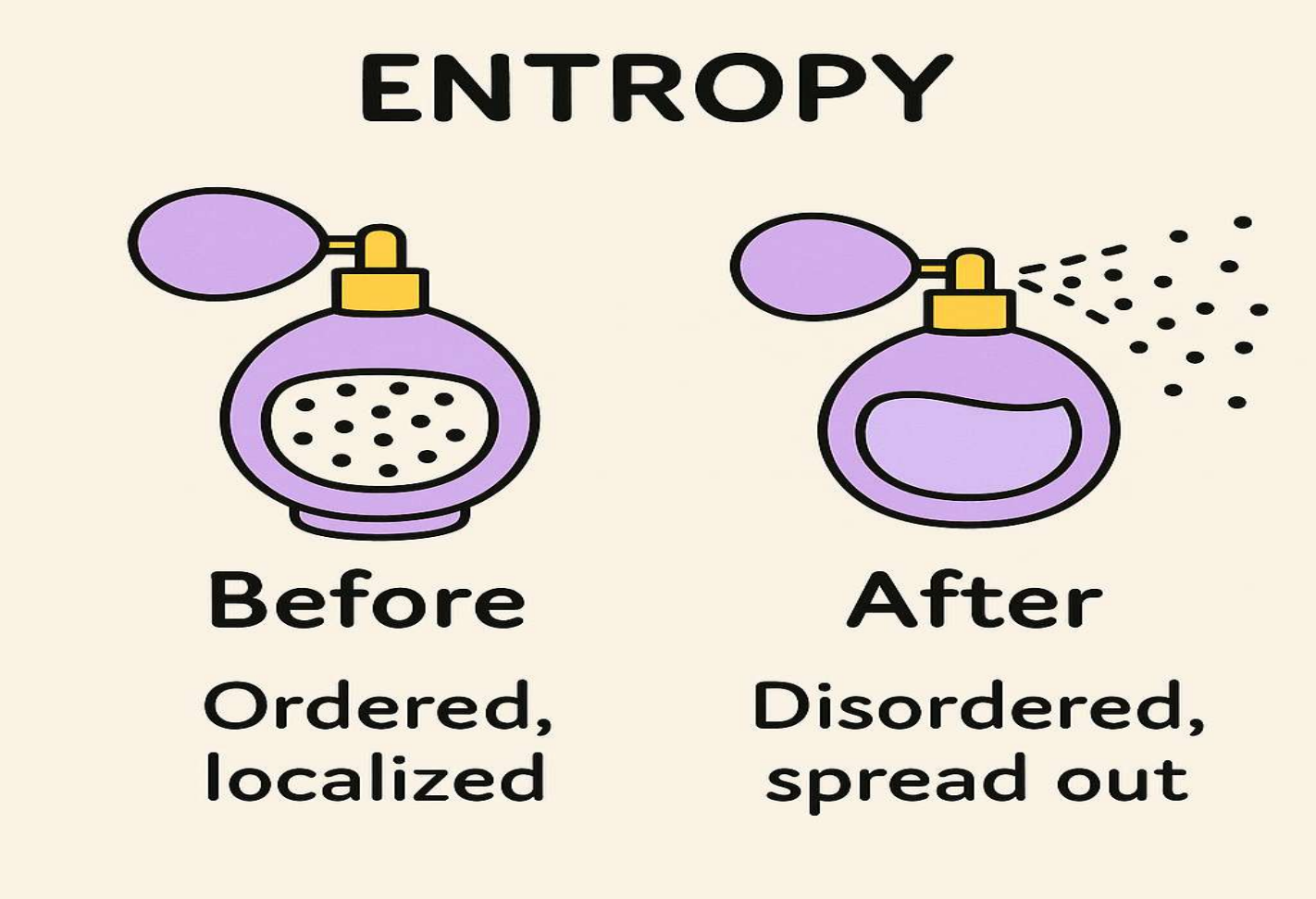
Sugar Cube and Entropy: A Simple Analogy (3/3)



Perfume Spraying in a Room (1/3)



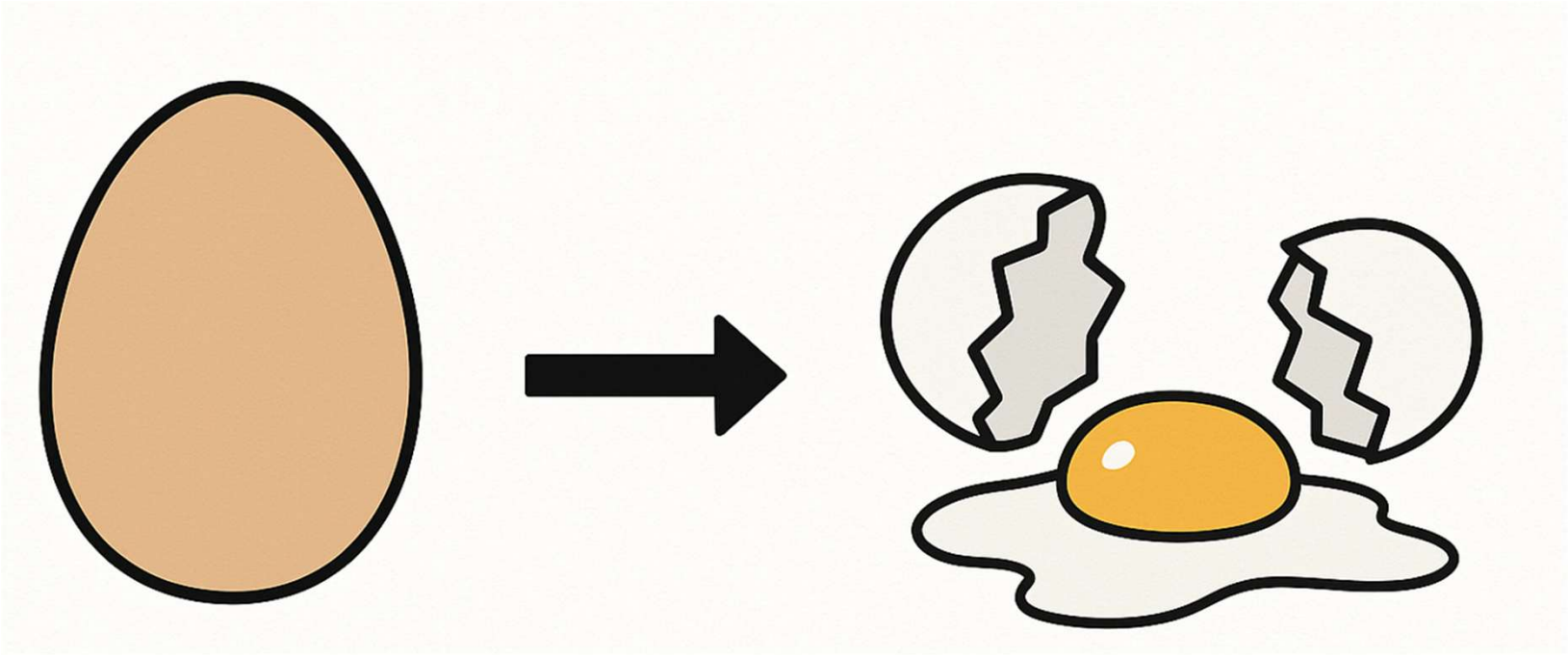
Perfume Spraying in a Room (2/3)



Perfume Spraying in a Room (3/3)

- **Before:** Perfume is contained in a bottle (ordered, localized).
- **After spraying:** The molecules spread and fill the room randomly (disordered).
- This is a **classic entropy increase** — gas molecules naturally spread out to occupy available space.

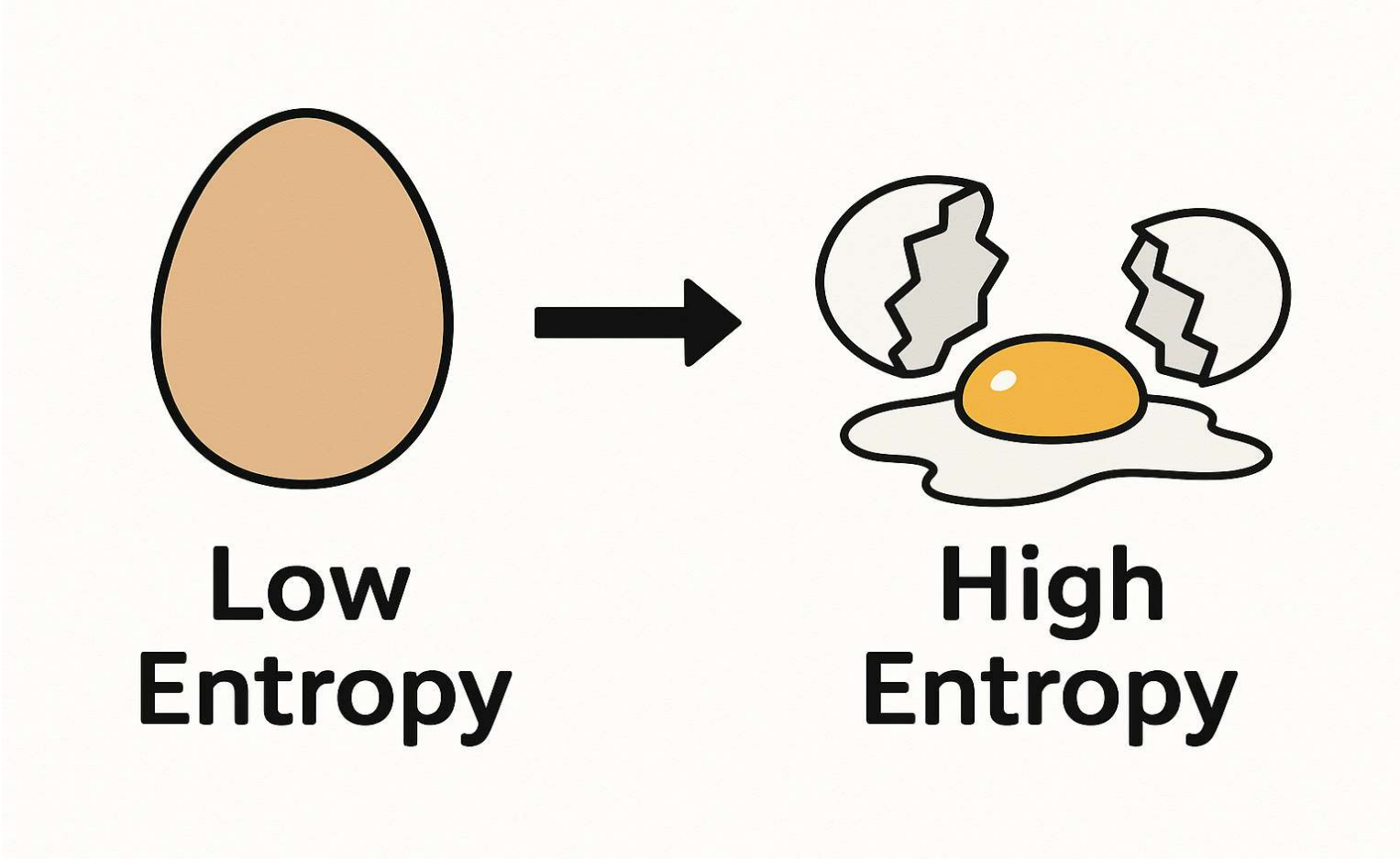
Breaking an Egg (1/3)



Breaking an Egg (2/3)

- **Before**: The egg is whole — yolk and white are neatly contained inside the shell (low entropy, ordered).
- **After**: The egg is cracked — contents are spread out or scrambled, mixed (high entropy, disordered).
- It's irreversible — just like increasing entropy.

Breaking an Egg (3/3)

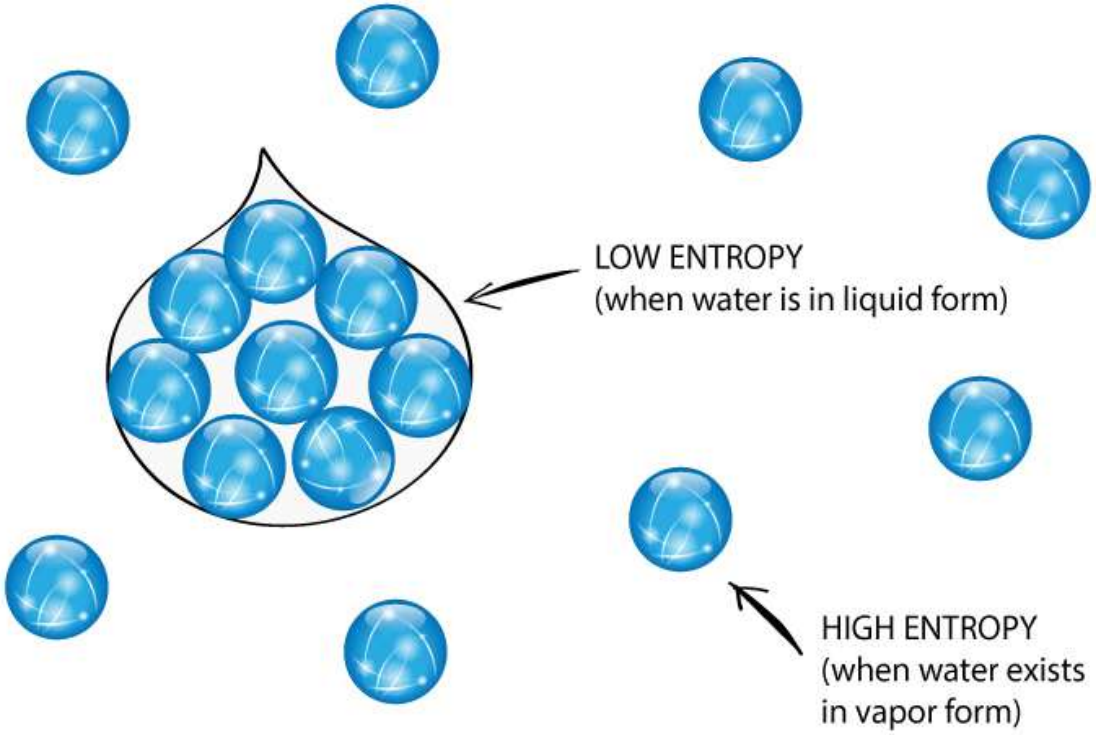


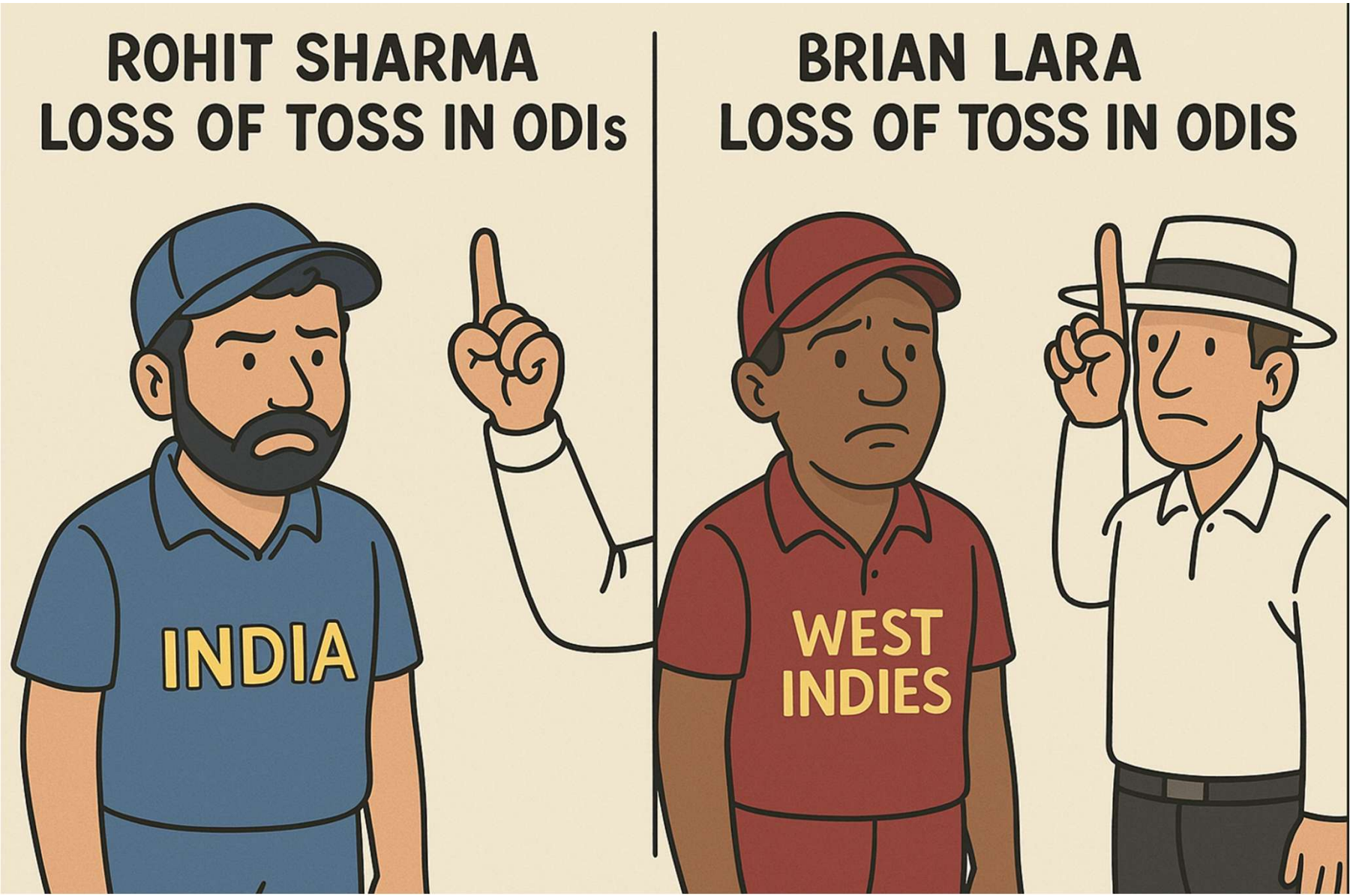
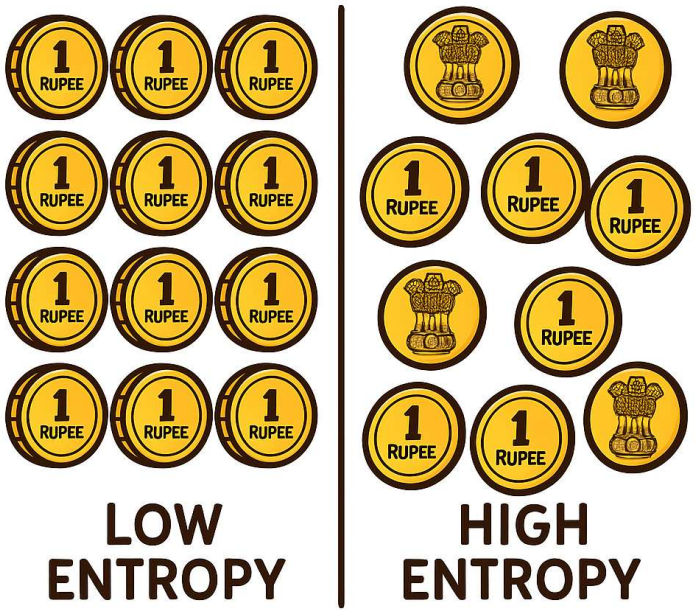
Ice Melting Example (1/3)

Ice Melting Example (2/3)

- Ice (solid water) is **structured and ordered** → low entropy
- When it melts into water, molecules move freely → **higher entropy**

Ice Melting Example (3/3)




ROHIT SHARMA
LOSS OF TOSS IN ODIs
INDIA
BRIAN LARA
LOSS OF TOSS IN ODIS
WEST INDIES

Tossing Indian Coins by Rohit Sharma Vs Other Captains